Samantha Zane Sanchez

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Education

Texas A&M University — College Station, TX
Master of Science in Visualization — *Aug. 2026 – 2028*

University of North Texas — Denton, TX
Bachelor's in New Media with a minor in Computer Science & Art History — *Aug. 2021 – Dec. 2024*

Institute of American University — Aix-en-Provence, France
Coursework in Art History — *June 2023 – July 2023*

Experience

Heidelberg Materials — May 2026 – Present
Application Developer & Digital Growth Intern — *Irving, TX*
- Designed & developed automated workflows, internal productivity apps & AI-enabled solutions using Microsoft Power Platform to address business needs & reduce inefficiencies in data retrieval by 25%.
- Contributed to iterative software development through Git-based version control, branch-based development, and Agile sprint workflows tracked in Jira.
- Built a centralized repository for organizing stakeholder requirements, operational data, & workflow insights to support scalable digital solutions through research & prototyping.

Ducki (Startup) ducki.ai — May 2024 – Aug 2025
Software Engineering Intern — *Austin, TX*
- Implementing retrieval-augmented generation (RAG) systems for an AI tutoring platform focused on personalized learning.
- Developed accessible interfaces, working across front-end, cloud, & AI systems to prototype & implement product features.
- Contributed to application performance improvements using Next.js & AWS.
- Collaborated on features including OAuth authentication, voice-to-text interaction, PDF processing, and personalized learning sessions.

Full Stack Software Development Program — Aug 2025 - February 2026
Software Developer — *University of Texas Austin*
- Selected into advanced developer cohort (top 15%) of program, focusing on scalable backend architectures, API design, & cloud-based deployment using the MERN stack.
- Developed full-stack applications using React, Node.js, Express, MongoDB, & RESTful APIs.
- Engineering RESTful APIs, database schemas, & background processing pipelines supporting 100K+ user records, improving backend efficiency by 45% through caching, aggregation, & async data handling.

5p Marketing — Aug 2022 – Aug 2025
Web Developer & Assistant Stage Manager — *Fort Worth, TX*
- Designed & developed website for Fort Worth Michelada Festival, increasing web traffic by 15% tracked through Google Analytics.
- B2B experience coordinating technical production with stakeholders & vendors to manage food & beverage logistics, sponsorships, & community engagement, optimizing event branding and audience experience.

Leadership

President of New Media Cooperative — Aug 2022 – Dec 2024
Lead Web Developer — *Denton, TX*
- Sole website Developer for my New Media Cooperative website with use of HTML, CSS, & Javascript; the site showcased the orgs mission & projects, contributing to a 30% increase in membership.
- Managed digital production with asset preparation, metadata handling, video conversion, audio preparation, and technical output configuration for events.

Society of Hispanic Professional Engineers (SHPE), Texas A&M University — Aug 2026 – Present
SHPEtinas & Technical Affairs — Website & Application Development — *College Station, TX*
- Participate in TAMU SHPE's app and web development teams, attending technical meetings and preparing development environments by installing required libraries and dependencies.
- Support upcoming web development initiatives focused on improving SHPE's digital interfaces and member experience.
- Participate in technical outreach and member engagement activities, such as an NFC workshop.

Selected Projects

Espresso Machine Fluid Simulation | *Blender, FLIP, 3D Simulation*
- Developed an espresso fluid simulation using Blender's FLIP workflow, configuring domain, inflow, collision, and liquid mesh systems.
- Iterated on simulation parameters, caching, and meshing to achieve controlled fluid motion and realistic surface detail.

Book Club App | *JSX, React, Tailwind, Vite, VS Code*
- Developed a data-driven web application for tracking books, reading progress, and user interactions.
- Implemented responsive React components and dynamic application state using a component-based architecture.

Touch Responsive Arduino LEDs | *Arduino, Blender, LEDs*
- Developed a touch-responsive LED sculpture integrating physical interaction with Blender-based data visualization.

Technical Skills

Programming: Python, C/C++, JavaScript, TypeScript, SQL, Java, Kotlin, HTML/CSS
3D & VFX: Blender, Maya, Mantaflow FLIP, Houdini, GLSL, ShaderToy, Unreal Engine, Unity, 3ds Max, Touchdesigner
Web & Software: React, Next.js, Node.js, Express, REST APIs, FastAPI, MongoDB
Tools: Git, GitHub, VS Code, Jira, Figma, AWS, Power Apps, Power Automate, Firebase, Adobe, Microsoft Suite/Power Apps
Certifications: UT Full Stack Software Development, CCNA, AWS Lambda Foundations, IBM AI Fundamentals